

- Time: from 2nd semester
- Tools: Unity / Fork / GitLab / Pro Create / FMOD / Miro / Logic Pro
- Team: Mai Rutkowski, Marika Ayumi Zobel, Jinzi Niu, Dohyun Lee, Ian Steiner
- My Role: Cave Level Programmer / Assets Artist / Level and Sound Designer



HARP

2D Music Puzzle Game

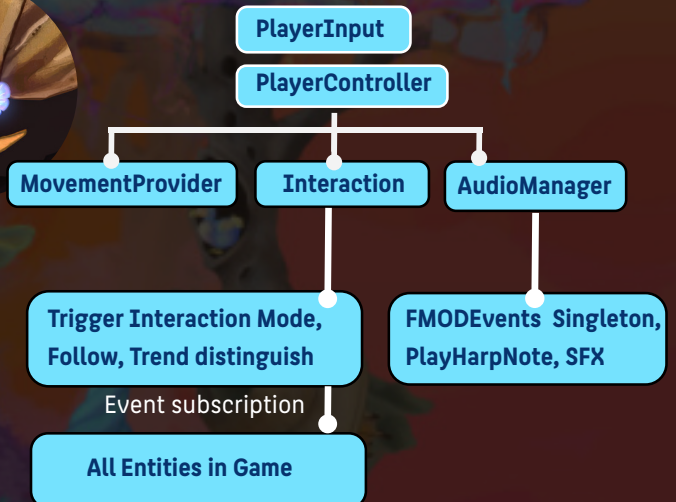
HARP is a musical puzzle game in which the player controls a creature named HARP and explores an organic world.

From movement to communication with other creatures, every interaction is built around listening to and identifying musical tones. The player begins at the top of the clouds, then travels downward through the world, eventually reaching its deepest layer.

Through this journey, the player witnesses the birth and death of HARP's life. The game uses music and visual design to emphasize a narrative about the cycle of life.



System Design



Level Design

The core puzzle mechanic of HARP is based on musical tones. Since not every player can easily distinguish pitch, we designed the levels with a gradual learning curve.



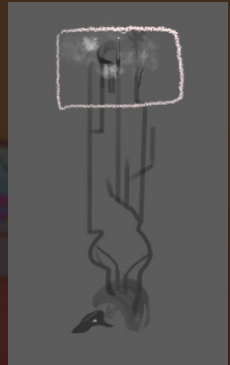
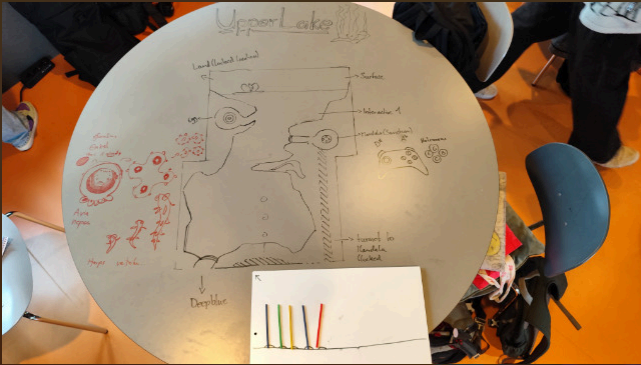
In the Sky level, players only need to recognize and repeat basic scales. In the Jungle level, they also need to identify how the same pitch sounds when played by different instruments.

In the Cave level, players use auditory memory to repeat sequences of tones. As praise, the world responds with a music and animation performance.



The final Big Harp level brings these skills together. Players rely on what they have learned to imitate melodic changes and create a musical ensemble, bringing the journey to its conclusion.

Graybox Design / first fakescreenshot



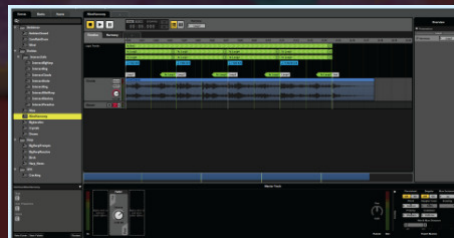
Prototype

The eight notes are divided into two groups:
Low register notes (0–3): C, E $\flat$, G $\flat$, A. These notes are also used for the Random sound effects. High register notes (4–7): D $\flat$, E, G, B $\flat$

A rising sequence moves the player upward, a falling sequence moves the player downward, and three repeated notes allow the player to hover.

In interaction mode, the melodic direction – Rising, Falling, or Steady – determines the outcome of the player's interaction with other creatures.

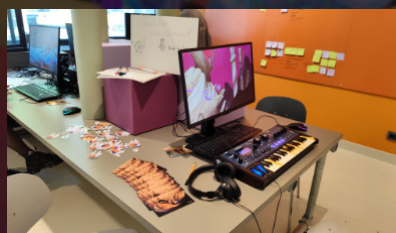
The notes are also divided into four visual-effect pairs: C/D $\flat$, E $\flat$ /E, G $\flat$ /G, and A/B $\flat$. Each pair triggers a different air-hole particle effect.



Feedback & Reflections



Team MajiDoia, 06.2025



POSITIVE:

The game's visual style is distinctive and received a lot of positive feedback. I am especially grateful to Mai for her exploration of materials and colors, which taught me a great deal throughout the project. As an experimental game, HARP also pushed us beyond our comfort zone. Its movement system encourages players to actively think about how to progress through the game.

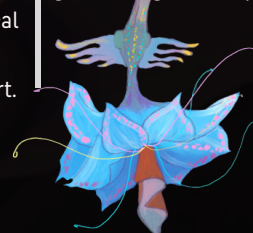
In addition, we used piano keys and an alternative controller to create a more physical experience of "making music" through play. Thanks to Ian, our controller and music expert.



NEGATIVE:

The game was not very accessible for most players, and there were many areas where the communication could have been improved.

In my own level, I removed the Interaction Mode with the intention of making the experience smoother. However, this had the opposite effect: many players did not understand how to continue. This became a major learning experience for me. It showed me how important player-friendly mechanics and consistent game design are, especially in an experimental project.



Hypnagogia

Hypnagogia (SAP) is a stylized 3D Survival Horror Game. Players are dropped onto a seemingly abandoned island, and to stay alive they need to recharge their battery at one of the six charging stations – each of which can only be used once.

Each time a player interacts with a charging station, the clock at the escape door ticks up, and once the counter reaches four, the escape door opens. How long does it take you to escape?



This game's core concept was inspired by the SURVIVAL structure in the LUNA Framework.



- Time: from 4th semester, 16 Days Games Jam
- Tools: Unreal Engine 5.7.4. / P4V / MAYA 3D / Substance Painter
- Team JELMI: Ian Steiner, Mai Rutkowski, Elena Woppmann, Lila Walter, Jinzi Niu
- My Role: UI Designer/ Programmer / Sound Designer / 3D Model

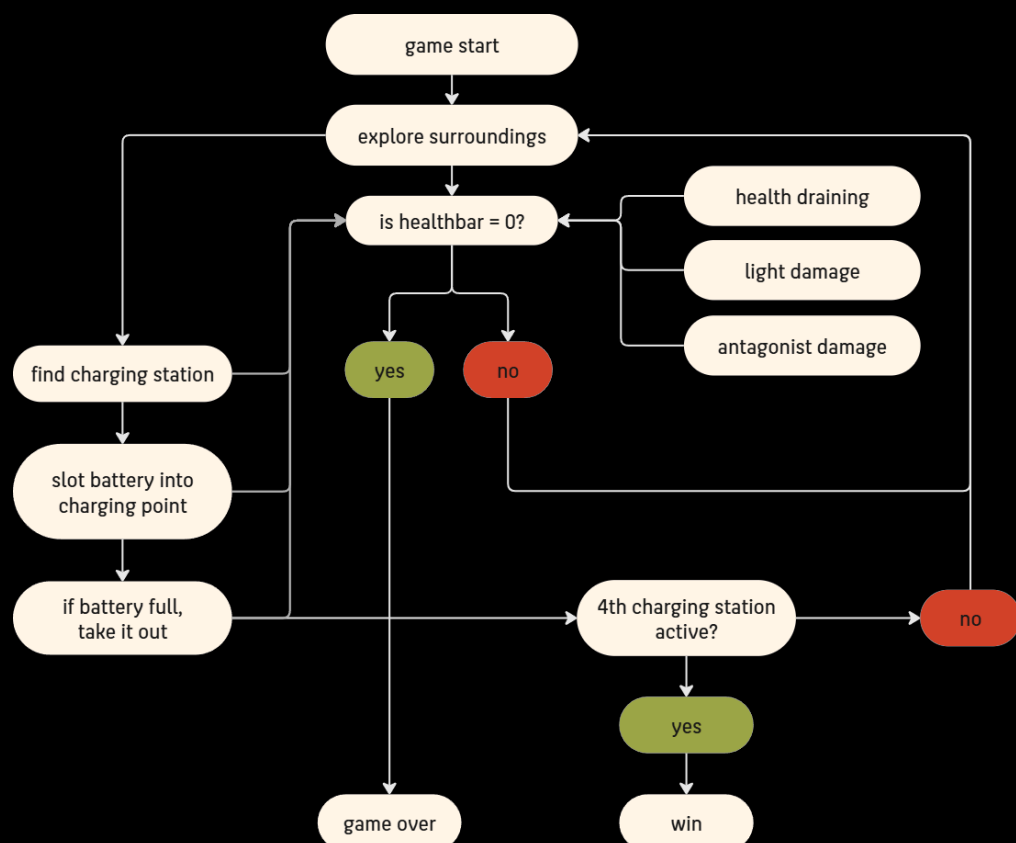


The game can be played with a joystick and an NFC reader – in that way, players can really feel like they're slotting a battery into one of our stations. Furthermore, SAP can be controlled using a keyboard and mouse.

Core Game Loop



Lighting quickly emerged as a powerful affordance in our design process. We ultimately decided to tie together light, batteries, and health into a single gameplay loop, creating a survival experience centered around managing a limited power supply.





Key Design Decisions



The island is inhabited by an invisible entity that continuously threatens the player. The entity begins pursuing the player whenever the flashlight is turned on or when the player enters a certain detection radius. At all other times, it follows a predefined patrol route across the island.

To implement this behavior, we built an AI Controller alongside a Blackboard and Behavior Tree.

1. Charging Minigame = Commitment Window

Charging is more than simply interacting with a station. Players must complete a small marble maze while charging. This creates a classic horror-game commitment window: the player wants to run, but can't, forcing them to overcome the tension and finish charging.

2. Flashlight = Dual-Edged Resource

The flashlight helps players navigate faster, but also makes them easier to detect. Using it reduces navigation time while increasing the pressure during escapes and charging.

We originally designed the monster to prioritize destroying charging stations. This allowed players to intentionally use the flashlight to lure the monster away and create a strategic decision. After our first playtest, however, we followed our instructor's feedback and simplified the design to reduce the cognitive load.

3. Health = Battery

Whether health and battery should be separate resources was one of the bigger debates in our team. One idea was to split the UI into a battery bar and a health bar. In the end, we merged them into a single resource.

Because resource management is at the core of the survival experience, making battery equal health makes every decision more meaningful. Taking damage directly reduces the energy needed to survive, making both enemy encounters and the choice of whether to use the flashlight much more impactful.

Hypnagogia (SAP) Download Link



PRESS ANY BUTTON TO START



Feedback & Reflections

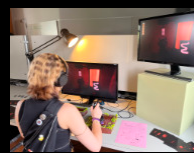


Team JELMI



POSITIVE:

The charging puzzle leverages 3D collision volumes aligned with sliding rails, creating an interaction that feels both intuitive and highly satisfying. Environmental audio—including thunder, whispering, and terrain-dependent footstep sounds—significantly enhances player immersion. The game's horror atmosphere is executed remarkably well, making the overall experience its greatest strength. In particular, the opening cinematic sequence is highly polished, immediately establishing a cinematic style of storytelling and drawing players into the world.



NEGATIVE:

Choosing joystick controls for both navigation and interaction was a bold design decision, and one we stand by. That said, joystick drift can negatively impact camera control, while the stick's inherent inertia may lead players to unintentionally hug walls during exploration. Additionally, the game's materials could be refined further. Striking a balance between a strong atmospheric presentation and sufficient visual readability remains an important area for improvement.



- Time: from 3rd semester
- Tools: Unreal Engine 5.6.1. / P4V / MAYA 3D / Substance Sampler / ProCreate / Blender
- Team DoJiSaLe: Leon Bublitz, Sara Samar Kaymaz, Jinzi Niu, Dohuyn Lee
- My Role: Technical Art / Programmer / Game Designer

FAR



Game Play Video

FAR is a stylized 3D Racing Game. The player controls the 'Floaty' to collect 10 Stars to finish the game. Unlike a normal racing game, where you drive on a track, in FAR the player can run on all surfaces -- especially on canyon and on the wall of towers.

This game has a special speed-storage system. The player's acceleration continuously accumulates, eventually creating the feeling of flying through the air, as if drifting across the galaxy.

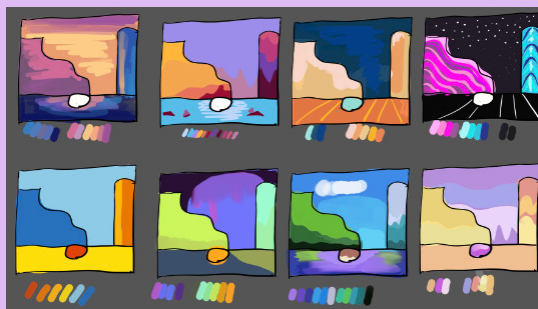


Prototype | Level Design



During the prototype phase, we defined a game structure centered on map design, supported by events that affect the player. We also developed several player abilities, including Break, Dash, Smash, and Wall Kick, which add more dynamism to the movement.

I explored many level design ideas, such as lava, bridges, and other structural elements. Although most of this work was later cut from the final project, the production process became a valuable exploration of level design.

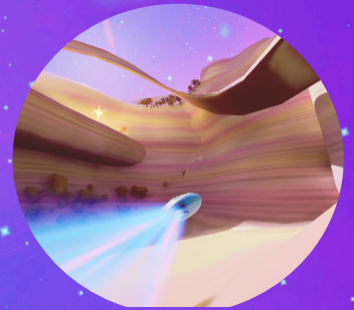


Through this process, we discovered the distinctive shape of a bone flute, which became an important visual reference. Based on team discussions, I designed several versions of "Floaty" with different silhouettes and surface finishes.



FAR Download Link:

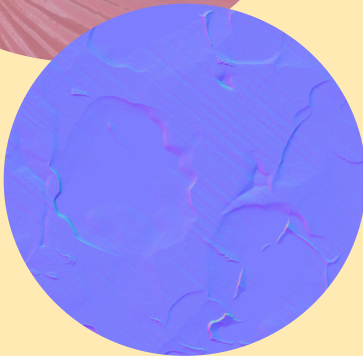
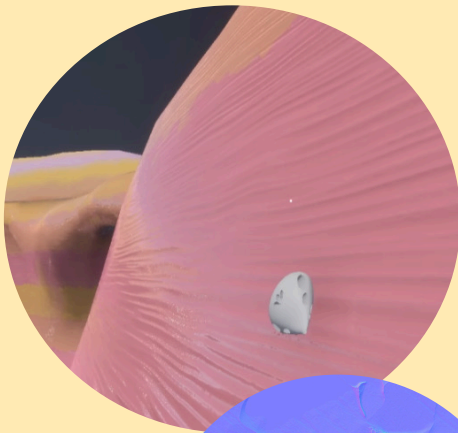
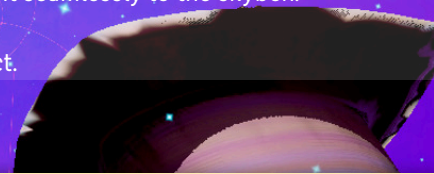




VFXs

The glowing stars, moon, and floating bubbles in the skybox were created with VFX. I originally made this tail flame effect for the player character UFO, and by adjusting its movement speed and parameters, I was able to adapt it seamlessly to the skybox.

And the visual effect of collectibles is made with mesh VFX, aiming to produce a fancy laser effect.



Team DoJiSaLe

The canyon and its surface materials were created in Substance Sampler. By adjusting key parameters such as Subtract and rebaking maps like the normal map, I achieved a rounded, stylized ground surface with a cartoon-like look.



Feedback & Reflections



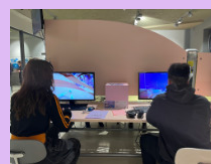
FIVE:HIVE Stream and Team Feedback



POSITIVE:

The visual design was praised by many players, and the flying sensation after acceleration proved to be genuinely exciting. A flying player character with a clear visual identity. This process allowed us to first create an enjoyable “toy” and then expand it into a game.

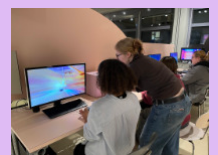
Our workflow was carefully structured throughout the project. Weekly progress was clearly documented, and the team maintained a clear division of tasks.



NEGATIVE:

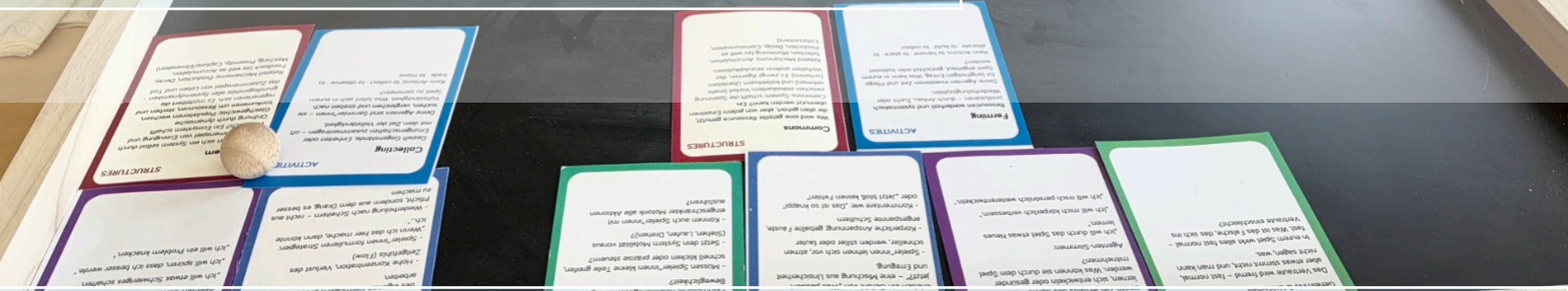
The camera control was not intuitive enough for players who were unfamiliar with racing games, and some players misread how the camera was meant to behave. Entering the flying state was also not achievable for every player, which showed that this mechanic should have been simpler and more controllable.

Looking back, some of our design decisions still left room for further discussion and iteration.



LUNA Framework

- Time: 04/2025 – 06/2026
- Tools: GitLab / Claude Code / Visual Studio Code / Miro / Analog
- Team: European Game Science Lab of Prof. Thomas Bremer
- My Role: Working Student – Stack content production / feedback implementation / Tabula template design



The LUNA Framework is a card-based design methodology that visualizes and structures the entire game design process.

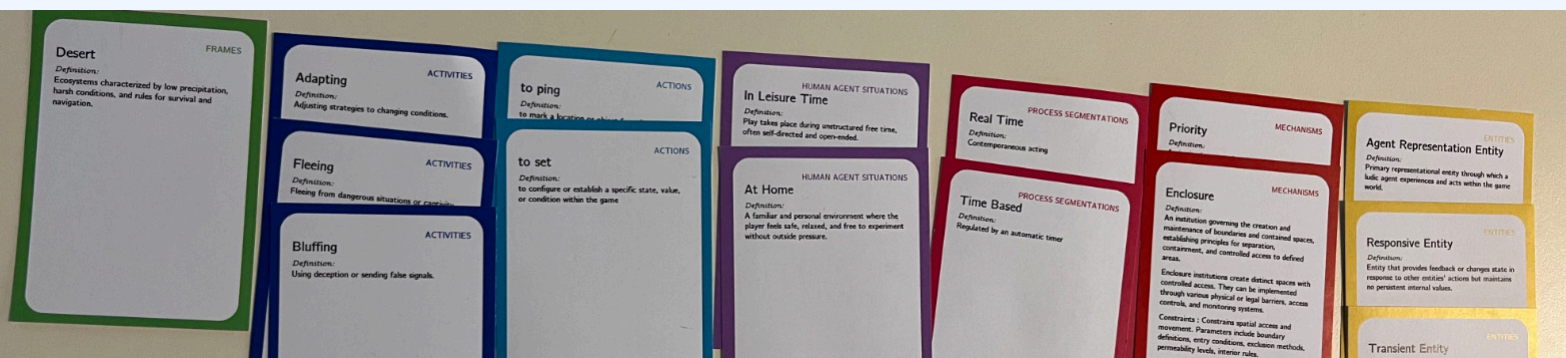
It is divided into three core areas: Agent (player experience), System (game systems), and Form (presentation & interface).



In addition, I co-designed the Tabula templates used during production. These templates organize the attributes of game entities, mechanics, and player actions into a shared documentation format, improving team communication and helping translate abstract design concepts into production-ready specifications.

My contribution focused on the Agent framework. I designed the Situation Stack, defining the contexts in which players engage with a game, and contributed to the Action & Activity Stack, categorizing player actions into higher-level gameplay systems. We also collected transcripts from thousands of videos as training data for an LLM, with the goal of building an AI agent for game design.

The LUNA Framework has already been applied in practice. Our 3D game FAR was developed within the Racing Frame, with movement and adaptation as its core mechanics.



- Time: from the 1st semester
- Tools: Super Sculpey / PolyCam
- solo project

MAQUETTE

A revenge-driven character designed with a blend of steampunk and medieval aesthetics.

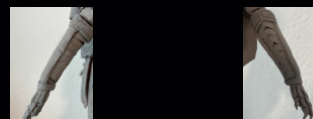
Born into royalty, Tyra · Moritz is the black sheep of the royal family, standing in sharp contrast to the values and expectations of her noble lineage.

After witnessing her lover being executed for their faith, Tyra swore to overthrow the Moritz Dynasty and end its reign of terror once and for all.

Her body has been extensively modified with mechanical components, making her a half-human, half-machine warrior. These augmentations are not merely cosmetic, but also serve as the source of her combat power.

Tyra's left arm is fully mechanical and functions as a versatile combat tool. It can transform into various forms, including a grappling hook and other utility devices.

The model incorporates wire, nylon stockings, decorative stamp patterns, and beads repurposed from hair accessories, each chosen to simulate specific textures and ornamental details.



The mechanical arm was particularly challenging. To achieve the layered metal plating and segmented structure, I cut and shaped pieces from an aluminum beverage can.



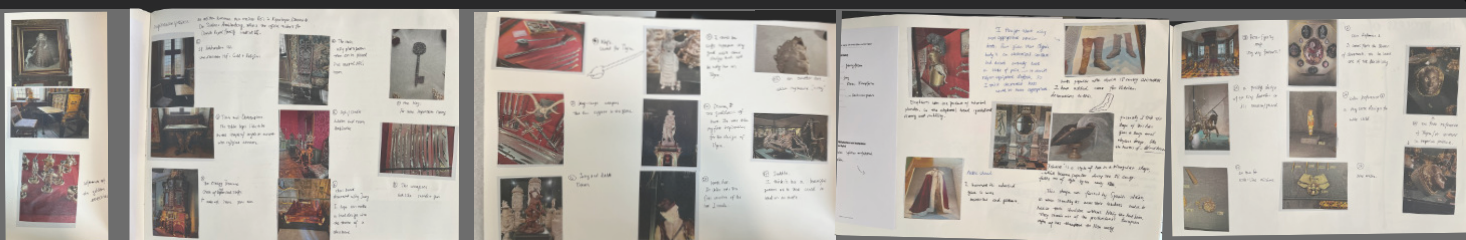
The Original Design Plan



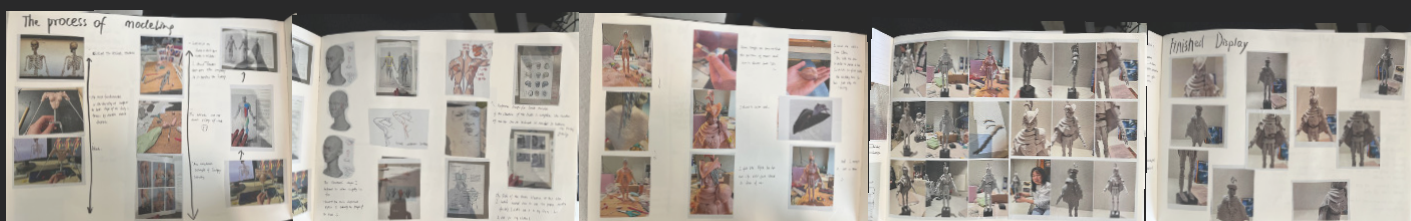
The final version

character name: Tyra Moritz

Trait: Shapeshifter - half Metal Human



Several visual elements of her design draw inspiration from the Amalienborg Museum in Copenhagen. The elephant motif, traditionally associated with power, prestige, and wealth, appears throughout her costume and equipment. The elaborate ornamental carvings and richly decorated surfaces reflect the extravagance and excess of the aristocracy of that era, reinforcing both her royal origins and the decadent world from which she emerged.



It took 5 weeks to lock down the design and setting, during which I also learned to build the base armature. Then 2 weeks of intensive sculpting. These are the steps and struggles from those weeks.